Jayhackers

University of Kansas

Team 15

White Team Intrusion Report

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Attempted Login Attempts (6:33AM)

We have continued to monitor all activity on our Windows and Linux authentication logs, which has shown indications of mass attempted logons. Notably, we've seen these logons targeting our Linux servers, which is shown in the figure below.

```
eb 21 07:32:19 www.team15.isucdc.com sshd[24924]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=49.10.219.164
eb 21 07:32:19 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:19] "GET /api/vehicles HTTP/1.1" 200 -
eb 21 07:32:20 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:20] "GET /api/vehicles HTTP/1.1" 200 -
eb 21 07:32:21 www.team15.isucdc.com sshd[24924]: Failed password for invalid user bob from 49.10.219.164 port 35462 ssh2
eb 21 07:32:21 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:21] "GET /api/vehicles HTTP/1.1" 200 -
eb 21 07:32:22 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:22] "GET /api/vehicles HTTP/1.1" 200 -
eb 21 07:32:23 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:23] "GET /api/vehicles HTTP/1.1" 200 -
eb 21 07:32:23 www.team15.isucdc.com unix_chkpwd[24933]: password check failed for user (cdc)
eb 21 07:32:23 www.team15.isucdc.com sshd[24930]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=49.10.219.164  user=
dc
eb 21 07:32:23 www.team15.isucdc.com sshd[24930]: pam_sss(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=49.10.219.164 user=cd
eb 21 07:32:23 www.team15.isucdc.com sshd[24930]: pam_sss(sshd:auth): received for user cdc: 10 (User not known to the underlying authentication module)
eb 21 07:32:24 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:24] "GET /api/vehicles HTTP/1.1" 200 -
eb 21 07:32:24 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:24] "GET / HTTP/1.1" 200 -
eb 21 07:32:24 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:24] "GET /static/css/style.css HTTP/1.1" 304 -
eb 21 07:32:24 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:24] "GET /static/images/Logo.png HTTP/1.1" 304 -
eb 21 07:32:25 www.team15.isucdc.com sshd[24930]: Failed password for cdc from 49.10.219.164 port 47878 ssh2
eb 21 07:32:26 www.team15.isucdc.com unix_chkpwd[24940]: password check failed for user (cdc)
eb 21 07:32:26 www.team15.isucdc.com sshd[24938]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=49.10.219.164  user=
dc
b 21 07:32:26 www.team15.isucdc.com sshd[24938]: pam_sss(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=49.10.219.164 user=cd
b 21 07:32:26 www.team15.isucdc.com sshd[24938]: pam_sss(sshd:auth): received for user cdc: 10 (User not known to the underlying authentication module)
b 21 07:32:28 www.team15.isucdc.com sshd[24938]: Failed password for cdc from 49.10.219.164 port 47886 ssh2
eb 21 07:32:29 www.team15.isucdc.com unix_chkpwd[24946]: password check failed for user (cdc)
eb 21 07:32:29 www.team15.isucdc.com sshd[24944]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=49.10.219.164  user=
dc
eb 21 07:32:29 www.team15.isucdc.com sshd[24944]: pam_sss(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=49.10.219.164 user=cd
eb 21 07:32:29 www.team15.isucdc.com sshd[24944]: pam_sss(sshd:auth): received for user cdc: 10 (User not known to the underlying authentication module)
eb 21 07:32:30 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:30] "GET /tickets HTTP/1.1" 200 -
eb 21 07:32:30 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:30] "GET /static/css/style.css HTTP/1.1" 304 -
eb 21 07:32:30 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:30] "GET /static/js/tickets.js HTTP/1.1" 304 -
eb 21 07:32:30 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:30] "GET /get_tickets HTTP/1.1" 200 -
eb 21 07:32:31 www.team15.isucdc.com sshd[24944]: Failed password for cdc from 49.10.219.164 port 47898 ssh2
eb 21 07:32:32 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:32] "GET /buy_ticket?check=on HTTP/1.1" 200 -
eb 21 07:32:32 www.team15.isucdc.com unix_chkpwd[24955]: password check failed for user (root)
eb 21 07:32:32 www.team15.isucdc.com sshd[24951]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=49.10.219.164  user=
oot
eb 21 07:32:32 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:32] "GET /static/css/style.css HTTP/1.1" 304 -
eb 21 07:32:32 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:32] "GET /static/js/tickets.js HTTP/1.1" 304 -
eb 21 07:32:33 www.team15.isucdc.com flask[1055]: 199.100.16.101 - - [21/Feb/2026 07:32:33] "GET /get_tickets HTTP/1.1" 200 -
eb 21 07:32:34 www.team15.isucdc.com sshd[24951]: Failed password for root from 49.10.219.164 port 47910 ssh2
eb 21 07:32:34 www.team15.isucdc.com sshd[24958]: Invalid user scrat from 49.10.219.164 port 33632
eb 21 07:32:34 www.team15.isucdc.com sshd[24958]: pam_unix(sshd:auth): check pass; user unknown
eb 21 07:32:34 www.team15.isucdc.com sshd[24958]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=49.10.219.164
eb 21 07:32:36 www.team15.isucdc.com sshd[24958]: Failed password for invalid user scrat from 49.10.219.164 port 33632 ssh2
eb 21 07:32:37 www.team15.isucdc.com sshd[24961]: Invalid user scrat from 49.10.219.164 port 33640
eb 21 07:32:37 www.team15.isucdc.com sshd[24961]: pam_unix(sshd:auth): check pass; user unknown
eb 21 07:32:37 www.team15.isucdc.com sshd[24961]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=49.10.219.164
eb 21 07:32:40 www.team15.isucdc.com sshd[24961]: Failed password for invalid user scrat from 49.10.219.164 port 33640 ssh2
```

Due to the fact that the logins have been failing to authenticate, we do not currently view this situation as being immediately threatening. Although some authentication attempts have been successful, we have determined that the successful login attempts have been legitimate users. Some recommended mitigation would be to block the attempting IP. A potential mitigation strategy would be to automatic block offending IP addresses for a short time via a tool such as Fail2Ban.

Tickets Forged (7:21AM)

As of 7:21AM, we have determined that an attacker was able to successfully forge a ticket in our ticketing system. This vulnerability originated from the tickets server not authenticating requests to sign a ticket as having come from the backend server. Our temporary workaround to mitigate this, locking the maximum ticket duration, is not sufficient, and contained an implementation bug that caused the mitigation to be ineffective. A complete solution to this problem would be to introduce a shared secret/MAC that would allow the tickets server to authenticate requests before signing them, but this was deemed too time-consuming for our team to implement in the present, leading to this compromise.

Looking at logs, there have been no ticket-related requests to the backend service since competition began, proving that the backend service was bypassed in creating the forged ticket. The forged ticket was for the user pwned@isucdc.com. Evidence of this incident is displayed below.

```
info: tickets.Controllers.TicketsController[0]
      Issued ticket for pwned@isucdc.com
```

Below is shown our initial mitigation attempt which contains the bug, an incorrect subtraction.

```
TicketsController.cs - Notepad
File  Edit  Format  View  Help
    public class SignRequest
    {
        public string Subject { get; set; } = string.Empty;
        public long ValidFrom { get; set; }
        public long ValidUntil { get; set; }
    }

    [HttpPost("sign")]
    public async Task<IActionResult> Sign([FromBody] SignRequest req)
    {
        if (string.IsNullOrWhiteSpace(req.Subject)) return BadRequest();
        if (req.ValidUntil <= req.ValidFrom) return BadRequest();
        if ((req.ValidFrom - req.ValidUntil) > (60 * 60 * 24)) return BadRequest();

        try
        {
            var (payload, signature, alg, kid) = await _signer.SignAsync(req.Subject, r
            _logger.LogInformation("Issued ticket for {Subject}", req.Subject);
            return Ok(new { payload, signature, alg, kid });
        }
        catch (Exception ex)
        {
            _logger.LogError(ex, "Signing failed");
            return StatusCode(500);
Windows (CRLF)   Ln 34, Col 48   100%
User:   N/A   Computer:   tickets.team15.isucdc.com
```